

Vish Karthikeyan

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EDUCATION

**Stanford University**, School of Engineering, California (recipient of the *Dhirubhai Ambani scholarship*)

**Bachelor of Science Degree in Computer Science** (Theory of Computation track) | Minor in Public Policy Exp. June 2027

Tentative Master of Science Degree in Computer Science (Artificial Intelligence track) Exp. June 2028

*Activities and Societies:* Director of Outreach: Stanford Political Union ▪ Counselor and Section Leader: Bridge Peer Counseling Center

▪ Teacher: Stanford Splash ▪ Editor (Tech and Democracy): Stanford Political Review ▪ Mentor: Bechtel International Center

**CS Academy Erode**, India

All India Senior Secondary Certificate, Top percentile national distinction with 96% aggregate (100% in Computer Science) May 2023

All India Secondary School Exam, Top percentile national distinction with 98% aggregate May 2021

*Activities and Societies:* School Captain ▪ National Debate Finalist ▪ National Talent Search Examination Scholar

RELEVANT COURSEWORK

|                                            |                                            |                                              |
|--------------------------------------------|--------------------------------------------|----------------------------------------------|
| CS 229 (Machine Learning; graduate-level)  | CS 124 (Natural Language Processing)       | CS 109 (Probability for Computer Scientists) |
| CS 161 (Design and Analysis of Algorithms) | CS 254 (Complexity Theory; graduate-level) | CS 107 (Computer Systems)                    |
| CS 106B (Abstract Data Structures)         | MATH 104 (Applied Linear Algebra)          | CS 229Q (Quantum Computing; graduate level)  |

SELECTED PROJECTS

**Reducing Hallucinations in LLM Coding Agents via Uncertainty-Aware Decisions** (CS 229 Final Project) : [paper](#)

- Built GRAB, an XGBoost classifier that gates tool calls in LLM coding agents by predicting when to invoke information-gathering tools vs. proceed, framed as supervised classification over a repository-level hallucination feature vector
- Reduced hallucination rate from ~50–69% to 44% on SWE-bench Verified while maintaining optimal tool call efficiency
- Engineered a 9-feature repo-grounding and entropy-based input representation for repository-aware hallucination detection

**Provable Expressivity Limits of Transformer Architectures** (CS 254 Final Project): [report](#)

- Proved transformers are TC0-bounded circuits, establishing unconditional limits on sequential/compositional reasoning independent of scale
- Formalized inevitability of hallucination via diagonalization and compositionality failure via communication complexity in LLMs

**Bias Evaluation in LLM vs. Human Criminal Judgments** (Ongoing): Conducted matched-guise experiments comparing commercial LLMs (e.g., GPT-series, Claude) to human professionals on guilt/sentencing decisions using controlled case vignettes; analyzed linguistic cues for bias divergence in probabilistic vs. biological reasoning with Hugging Face Transformers and statistical metrics.

WORK EXPERIENCE

**Student Dining Worker**, Stanford Residential and Dining Enterprises Fall 2025 - current

- Hands-on food service operations experience at Stanford’s largest dining hall, serving 3000+ people a day in an extremely fast-paced environment
- Helped bridge communication between the student body and dining staff, contributing to a more student-centered campus dining experience

**Residential Assistant, Summer Session, Stanford University** Summer 2025

- Built an inclusive, supportive residential community for 300+ high school summer students by planning and leading 3+ programs a week
- Served as a key resource for student concerns and campus referrals; coordinated case management

**Teaching Assistant**, Frosh 101, Stanford University Fall 2024

- Inspired community-building conversations and activities amongst two classes of freshmen, shaping their college transitions
- Attended a quarter-long course on inclusive pedagogies and impacts of diversity, inclusion and sense of belonging on learning outcomes

**Reunion Homecoming Team Leader**, Stanford Alumni Association Fall 2024

- Led planning efforts, coordinated event logistics, and developed programming for over 7000+ alumni
- Recruited, trained, and managed a team of volunteers, delegating tasks and providing guidance to ensure efficient event execution

**Multiple leadership roles**, Associated Students of Stanford University 2023-24

- *Assistant Elections Commissioner:* Facilitated student government elections involving 18,000 voters, supervised data management and security; practiced constitutional hermeneutics and policy development
- *Senate Associate:* Participated in a leadership and advocacy workshop for six months; directed a team working on designing and implementing modifications to Stanford Dining
- *Neighborhood Council Representative:* Managed \$200,000 in funding across community building events for 800+ students, engineering various types of events and festivals; designed permanent community spaces for the Stanford Outdoor Enhancement Project

SKILLS AND INTERESTS

|           |                                                                                                                                                                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coding    | Python (including SciKit-Learn, PyTorch, TensorFlow, NumPy, Pandas and Hugging Face), C/C++                                                                                                                                                                                                                                        |
| Tools     | Git and version control, UNIX command-line, SQL (MySQL, SQLite), CSS, HTML, Javascript, RESTful APIs, Docker and containerization, Kubernetes, Agile development, Jupyter, supervised and unsupervised learning, data structures, algorithmic complexity, model evaluation and metrics, feature engineering and data preprocessing |
| Languages | English (native proficiency), Tamil (native proficiency), Spanish (elementary) and Hindi (elementary)                                                                                                                                                                                                                              |
| Interests | NLP, LLMs, computability and complexity theory, causal inference for public policy, political theory, mental health and peer counseling                                                                                                                                                                                            |